

AI-Driven Construction Contract Management System

Team X

Team Lead : Chak Lee Meng

Team Member : Tan Tian Chong

AI Hackathon 2025



- Manual processes cause inefficiencies and delays.
- Poor change order tracking and dispute management.
- Complexity of multi-party collaboration and regulatory compliance.

- Users: Contractors, Project Owners, Legal Teams
- Pain Points:
 - Slow contract approvals
 - High legal costs
 - Misaligned milestones & payments
- KPIs: 70% review time reduction, 50% better compliance, 30% fewer disputes

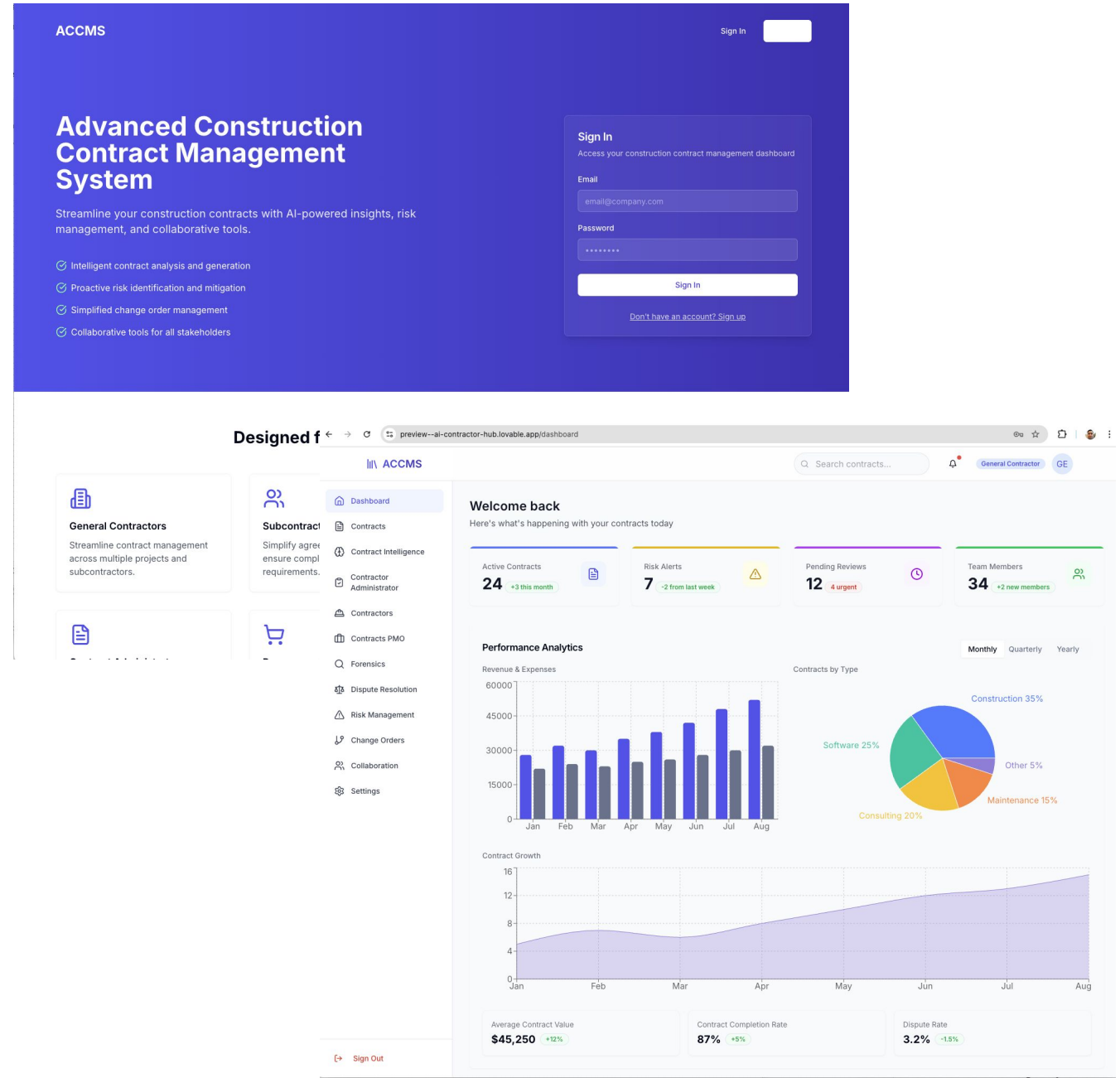
- Traditional CLMs lack integration with project systems.
- No AI-based clause risk detection.
- Manual milestone/payment tracking causes issues.

- \$1.4T lost yearly to mismanaged contracts
- \$12M avg loss per \$100M project
- Poor contract management = high dispute risk and cost overruns.

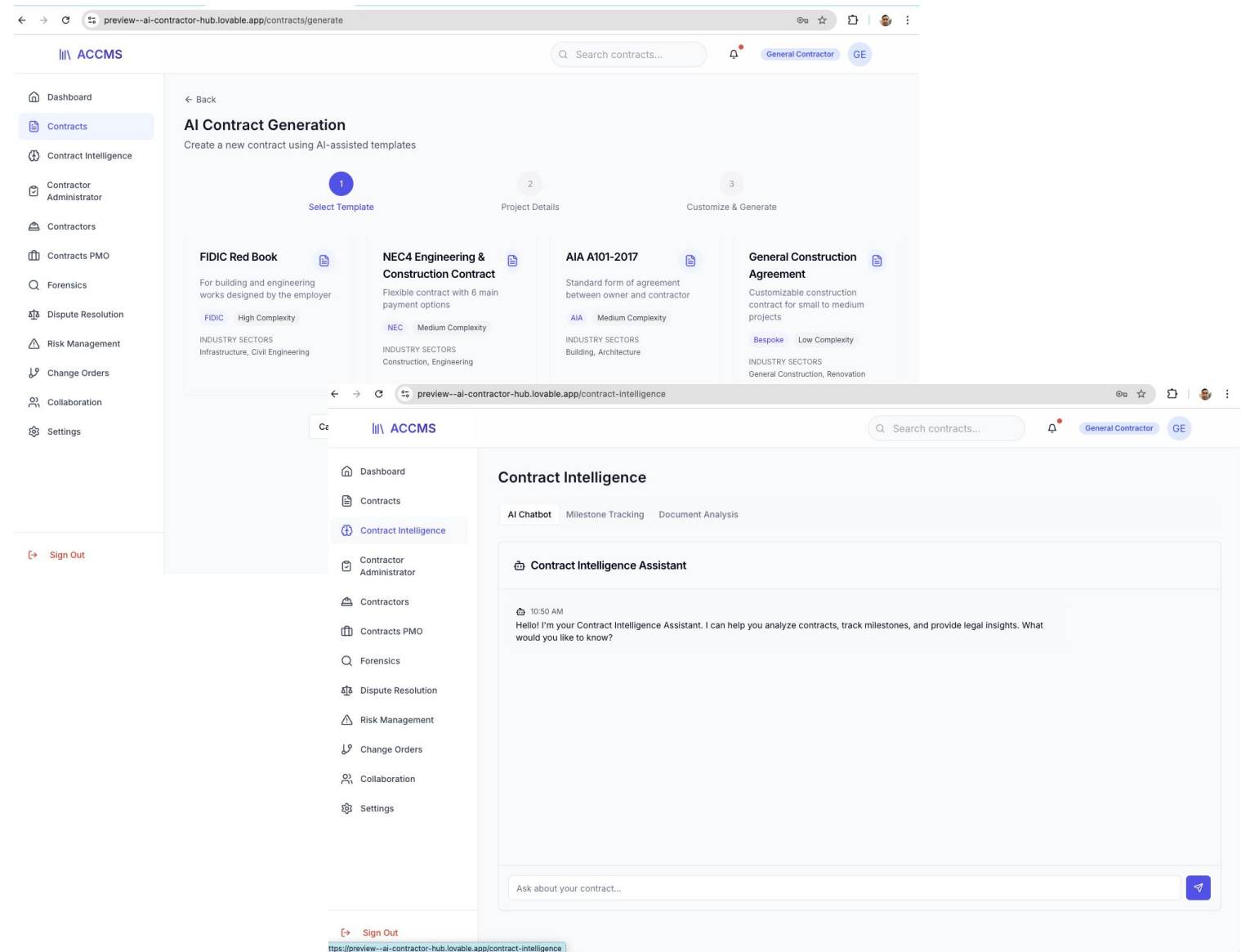
Scope of PoC & Target Outcomes



- Scope: AI-powered contract analysis, change tracking, compliance
- Outcomes:
 - Faster approvals
 - Proactive risk alerts
 - Smart milestone payment tracking



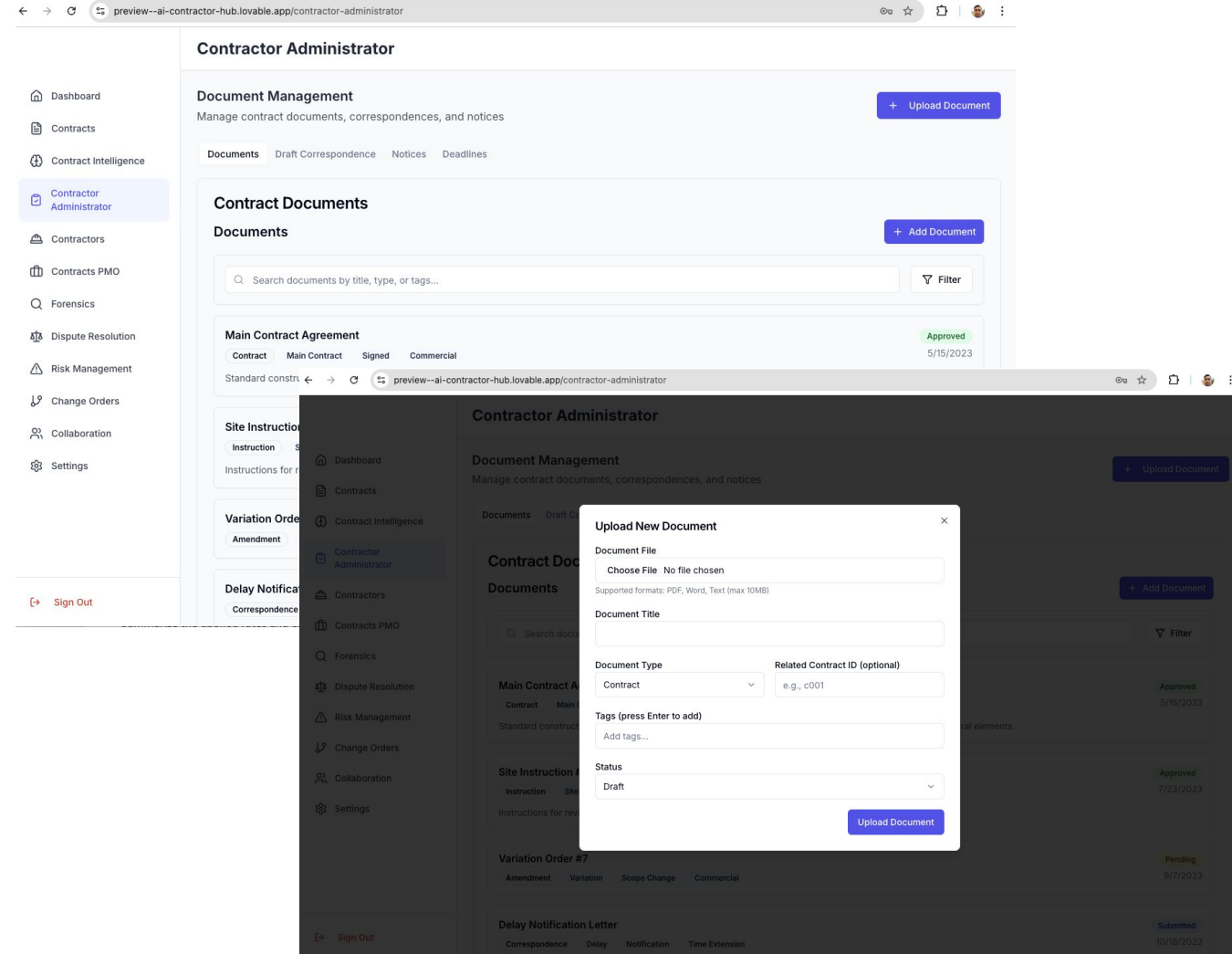
- GenAI: Drafting, clause extraction, NLP insights
- Agentic Workflow: Real-time monitoring of obligations
- RAG: Context-aware contract search & automation



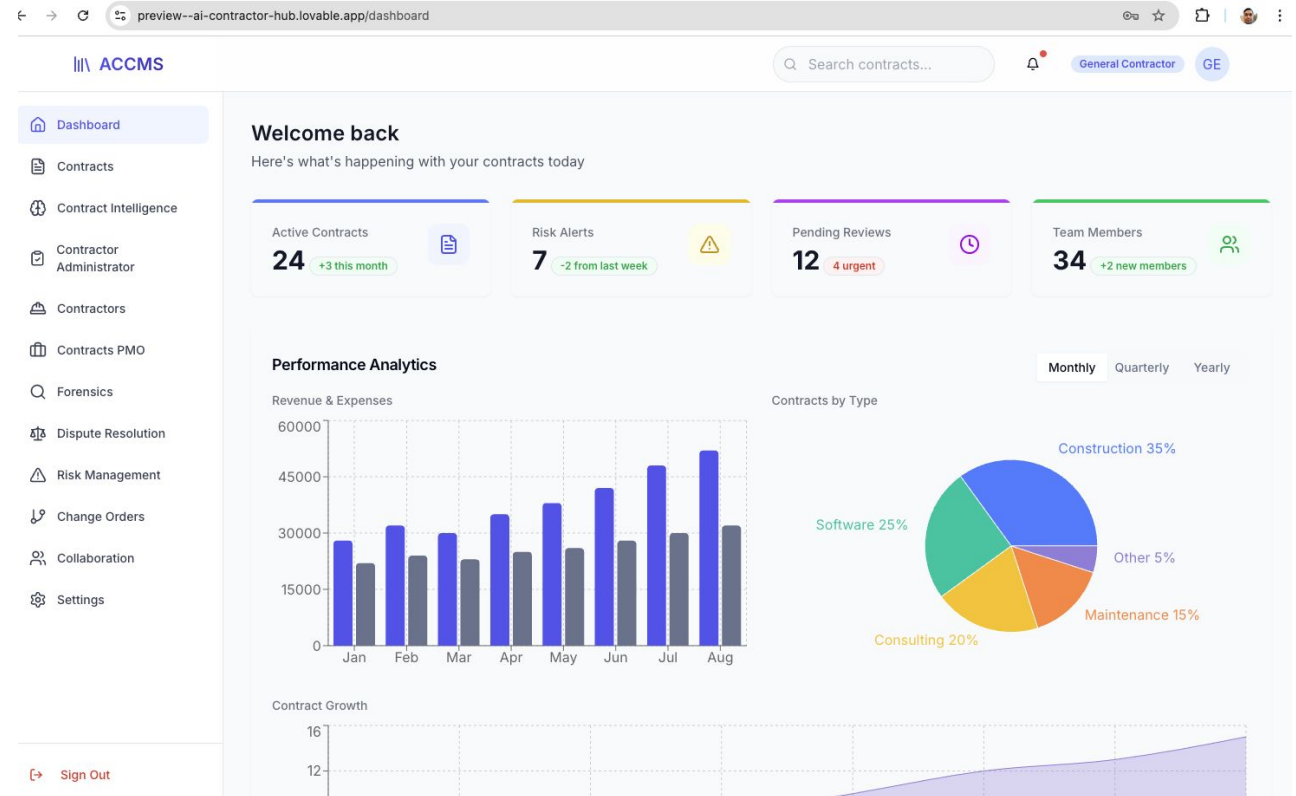
Data: PDFs, DWFs, DOCX
briefs, XLSX summaries

Steps:

- 1. Ingest & parse
- 2. NLP clause analysis
- 3. AI alerts & tracking
- 4. Smart contract execution



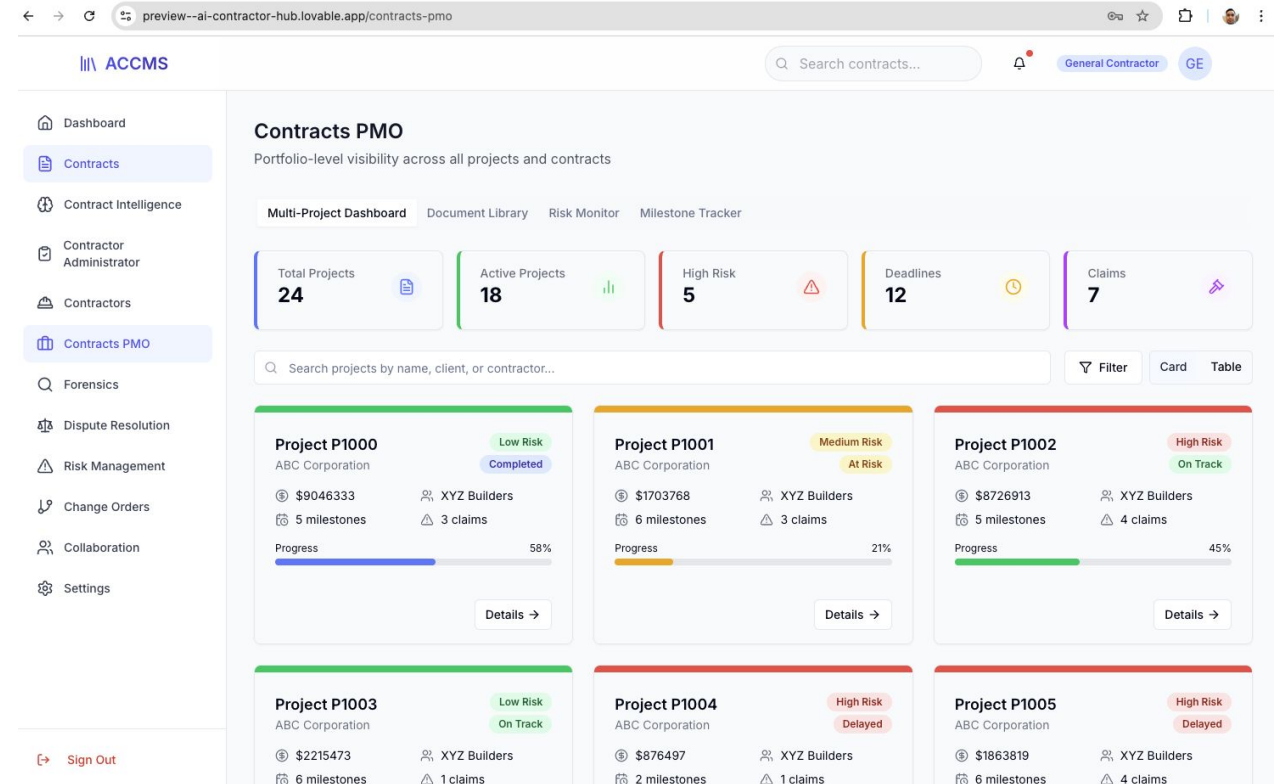
- GPT/LLM: Clause generation & redlining
- ArchiCad/BIM360: Real-time site sync
- Blockchain: Transparent audit trail
- RAG: Grounded contract Q&A



Tests, Results & Roadmap



- Validation: Partner firm pilots
- Results:
 - - 70% faster reviews
 - - 30% fewer disputes
 - - 40% better cost tracking
- Next: Global rollout, multilingual AI, deeper integrations



Thank you

